

MUSP-BENCH: AN ADVANCED MULTIMODAL MUSIC SCORE-PERFORMANCE UNDERSTANDING BENCHMARK

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ABSTRACT

Musicians commonly communicate music through two complementary modalities: scores and performances. Scores encode musical structure and intent, while performances realize them in sound. To investigate whether models can meaningfully engage with both modalities, we introduce **MuSP-Bench**, a human-authored benchmark of 490 questions targeting understanding across **Musical Scores** and **Performances**. The benchmark distinguishes itself through long-horizon, open-ended questions spanning score, performance, and interpretive reasoning across classical piano and orchestral works. We evaluate frontier multimodal large language models under multiple input conditions and formats. Our results show that these models struggle substantially to understand scores, while facing even greater challenges when reasoning about performance audio. The benchmark is at <https://musp.vaclis.net>.

1. INTRODUCTION

Musicians create and communicate musical ideas through both scores and performances. Scores encode compositional intent—pitch, rhythm, harmony, dynamics, articulation, and structure—while performance realizes them through timing, phrasing, voicing, technique, and timbre. Musicians continually move between these representations: reading, imagining, performing, listening, and returning to the score. Models aiming at human-level music understanding and serving real-world music applications, including education and artistic, practice should therefore be able to reason across both score and audio.

Existing benchmarks largely assess these modalities in isolation. MSU-Bench [1] evaluates ABC and image-based scores, but mostly through short-horizon or true/false questions. MusicTheoryBench [2], Ziqi-Eval [3], and WildScore [4] similarly focus on ABC or single-image excerpts and predominantly use multiple-choice evaluation. MuxQA [5] and SSMR-Bench [6] generate synthetic scores and questions on properties such as pitch, rhythm, chords, scales, and meter. While useful for symbolic understanding, these benchmarks pro-

vide limited coverage of higher-level musical interpretation. MuseBench [7] is one of the few benchmarks to jointly evaluate score and performance. However, its joint tasks mainly test whether tempo matches a reference, notes are omitted, or performance speed is consistent through true/false questions. As such, it does not address dimensions such as phrasing, voicing, rubato, technique, and expression. Benchmarks like HumMusQa [8], MUSE [9], CMI-Bench [10] test musical perception from audio, including rhythm, melody, harmony, tempo, and instrumentation. However, these benchmarks rarely assess the models’ understanding of the performance: how phrasing, dynamics, or expressive character shape a particular realization of a work.

Overall, existing benchmarks rarely operate at the level of complete musical works or the questions musicians ask of them: how phrases unfold, where interpretive and technical challenges arise, and how a performance relates to the written score. We therefore introduce a multimodal benchmark that combines local musical understanding with higher-level reasoning and real-world inquiries across complete works and across score and performance.

2. BENCHMARK

MuSP-Bench comprises 490 questions across 24 classical piano and orchestral works. Professional musicians authored questions requiring information from the score, performance, or both. 460 questions are open-ended within a predefined answer format, such as a pitch, chord progression, timestamp, measure range, or short phrase, and may allow multiple correct answers to reflect musical ambiguity; the remaining 30 have at least 7 options to choose from. Piano scores come from ASAP [11], corrected using aligned PianoCoRe MIDI [12]; orchestral scores and audio come from BSED [13]. Further details are available on the demo website.

Each question is annotated along four dimensions. **Modality** indicates whether the minimal required information comes from the score (S), performance (P), both (SP), or either (S/P). **Horizon** captures the smallest span over which evidence must be integrated: short (1–2 measures or less than 5 s), medium (3–16 measures or between 5–30 s), long (larger, noncontiguous, or piece-level), or any. Pieces average 10 score pages (1–40) and nearly 5 minutes (0:17–12:29). **Content** follows a hierarchy from *pitch*, through *temporal organization*, *orchestration*, and



performance realization, to melodic, harmonic, formal, interpretive, and contextual understanding. Finally, **Action** specifies the reasoning operation: identification, localization, quantification, comparison, inference, transcription, or explanation/summarization.

2.1 Question Overview

Representative score questions (S) assess harmonic functions, chord progressions, motifs, phrasing, interpretation, and voicing. Performance questions assess understanding of the played music (P or S/P) and its realization of the score (SP), including rubato, expressive emphasis, tone color, melodic variation, phrasing, voicing, fingering, and pedaling. Representative questions ask which note receives the strongest expressive emphasis, where a melody is varied, how a performer shapes a phrase, or which pitches constitute a secondary voice. MuSP-Bench also tests musical knowledge by asking models to identify a work’s composer, style, tonality, and title from musical content alone, without metadata, using either score or audio (S/P). Examples drawn from SP, S/P, and P modality:

- Consider the theme starting at 2:51. In which bar does the performer reach the subverted, silent climax of this theme, characterized by a significant slow-down in tempo and loudness? (SP)
- What would you say is the source for the main challenge for the pianist in this piece: polyphonic texture, complex polyrhythms, emotional depth, virtuosic arpeggios, large leaps, hand synchronization, or rapid scales? (S/P)
- How many F3 bass notes are played truly staccato in the performance before 0:08 (i.e. not elongated in any way)? (P)

3. EXPERIMENTS

We evaluate multimodal foundation models using the input modalities they support: GPT 5.6 Sol and Qwen 3.6 Plus on text and score images, Audio Flamingo Next on audio, and Muse Spark 1.2 and Qwen-Omni-Plus on text, image, and performance audio. Input depends on the modality. For score-only questions, we use ABC notation and PDF-scores converted to images; for performance-only questions, audio and MIDI rendered as text. For score-performance questions, we combine score images with audio and ABC notation with audio. A language-only baseline measures how far models can answer from prior knowledge of a work without receiving the score or performance, referred to as ‘context only’ in Table 1.

4. RESULTS

On score-only questions, GPT-5.6-Sol performs best with both ABC (50.6%) and images (26.1%). The substantial advantage of ABC suggests that models process structured symbolic notation more effectively than rasterized scores. Performance-only results likewise favor symbolic input: GPT reaches 59.4% with MIDI-as-text, compared with

Qwen3.5-Omni Plus’s 13.2% with audio. On combined SP questions, MuseSpark falls from 30.6% with ABC+MIDI to 6.9% with either Image+audio or ABC+audio, while Qwen3.5-Omni Plus remains weak across all three representations. The separated S/P subsets show that the value of each modality depends on the task. Symbolic input is strongest for general analytical questions, with GPT reaching 50.0% using MIDI-as-text and 44.4% using ABC. Audio and images perform well on global attributes: Qwen3.5-Omni Plus reaches 75.0% with audio on Tonality/style, while MuseSpark reaches 85.4% with images. Images also help with Composer/composition questions, where GPT and MuseSpark score 54.2% and 52.1%, respectively. Overall, GPT is the strongest and most consistent model, while visual and audio inputs offer particular benefits for stylistic and contextual recognition.

Table 1. Accuracy (%) by subset and input condition. The #Q column gives the number of eligible questions. A smaller gray bracketed number gives the number completed for an incomplete run. Bold marks the highest result in each row; underlining marks each model’s highest non-context input result within that subset, including ties. N/A denotes an unsupported input format or task. The original S/P questions are separated into general S/P, Tonality/style, and Composer/composition subsets. Composer/composition contains only Composer and Work questions. The final aggregates exclude Context-only baselines and weight the six subsets equally.

Subset	Input	#Q	GPT-5.6 Sol	Qwen3.5 Omni Plus	MuseSpark 1.2	Qwen-3.6 Plus	Audio Flamingo Next
Score only (S)	Context only	180	5.0	1.7	4.4	1.7	N/A
	ABC	180	50.6	11.7	<u>31.7</u>	11.1	N/A
	Image	180	26.1	<u>12.2</u>	15.0	<u>13.3</u>	N/A
Performance only (P)	Context only	106	5.7	4.7	5.7	3.8	3.8
	MIDI-as-text	106	59.4	11.3	<u>30.2</u>	<u>10.4</u>	N/A
	Audio	106	N/A	<u>13.2</u>	3.8	N/A	<u>0.9</u>
Score and performance (SP)	Context only	72	6.9	4.2	4.2	1.4	N/A
	ABC+MIDI	72	45.8	5.6	<u>30.6</u>	6.9	N/A
	Image+audio	72	N/A	5.6	6.9	N/A	N/A
General (S/P)	ABC+audio	72	N/A	4.2	6.9	N/A	N/A
	Context only	36	13.9	5.6	8.3	5.6	2.8
	ABC	36	44.4	<u>16.7</u>	33.3	11.1	N/A
	MIDI-as-text	36	50.0	13.9	<u>36.1</u>	11.1	N/A
	Image	36	19.4	<u>16.7</u>	11.1	19.4	N/A
Tonality/style (S/P)	Audio	36	N/A	<u>16.7</u>	11.1	N/A	<u>0.0</u>
	ABC	48	77.1	50.0	68.8	<u>50.0</u>	N/A
	MIDI-as-text	48	72.9	39.6	54.2	33.3	N/A
	Image	48	68.8	50.0	85.4	47.9	N/A
	Audio	48	N/A	<u>75.0</u>	20.8	N/A	<u>22.9</u>
Composer/composition (S/P)	ABC	48	35.4	12.5	43.8	8.3	N/A
	MIDI-as-text	48	27.1	0.0	<u>27.1</u>	0.0	N/A
	Image	48	54.2	14.6	<u>52.1</u>	18.8	N/A
	Audio	48	N/A	39.6	2.1	N/A	<u>0.0</u>
Aggregate	Context only	394	6.3	3.3	5.1	2.5	3.5[142]
	ABC	312	51.6	18.3	39.4	16.7	N/A
	MIDI-as-text	238	54.2	15.1	35.3	13.0	N/A
	Image	312	36.2	18.9	31.1	20.2	N/A
	Audio	238	N/A	31.5	8.0	N/A	5.0
	ABC+MIDI	72	45.8	5.6	30.6	6.9	N/A
	Image+audio	72	N/A	5.6	6.9	N/A	N/A
	ABC+audio	72	N/A	4.2	6.9	N/A	N/A
	Available-run mean	6 subsets	48.9	19.3	27.8	16.0	N/A
Aggregate	Top score per subset	6 subsets	56.2	27.0	44.3	19.8	N/A

5. CONCLUSION

We introduced MuSP-Bench, a human-authored benchmark for analytical and interpretive reasoning across musical scores and performances. Our results show that current models struggle with image-based scores and expressive performance audio. MuSP-Bench provides a foundation for measuring progress toward more meaningful musical understanding across modalities.

6. REFERENCES

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